香港資優教育學苑
The Hong Kong Academy for Gifted Education

IMO Training 2022-23 Phase II Level 1

Inequalities I

17th September 2022

There are two types of inequality problems.

Type 1: **Conditional inequalities (條件不等式)**

A conditional inequality is an inequality which is valid only when the variable(s) in the inequality takes certain values. (For example: $4x - 3 > 15 - 2x$ is valid only when $x > 3$.)

Ex. 1: Solve the inequality $\dfrac{x^2 - 9x + 11}{x^2 - 2x + 1} \geq 7$

Ex. 2: Solve the inequality $\sqrt{5(5-x)} > 5 - 2x$

Ex. 3: Solve $\frac{1}{x+1} + \frac{1}{x+4} > \frac{1}{x+2} + \frac{1}{x+3}$

Ex. 4: For what values of the variable x does the following inequality hold?

$$\frac{4x^2}{(1-\sqrt{1+2x})^2} < 2x + 9 \text{ ?}$$ (Second IMO (1960) No. 2)

Ex. 5: Determine all real numbers x which satisfy the inequality $\sqrt{3-x} - \sqrt{x+1} > \dfrac{1}{2}$.

(Fourth IMO (1962) No. 2)

Type 2: **Absolute Inequalities (絕對不等式)**

An absolute inequality is an inequality which is valid no matter what values the variables assume in the permitted number set.

For example: For any positive real numbers a, b, $a + b \geq 2\sqrt{ab}$

Proof: $(\sqrt{a} - \sqrt{b})^2 \geq 0$, hence $a + b \geq 2\sqrt{ab}$.

Ex. 6: For any real numbers a, b such that a>b, prove that $a^3 > b^3$.

Ex. 7: For any positive real numbers a, b and c, prove that $a^3 + b^3 + c^3 \geq 3abc$.

Ex. 8: Given that x, y, z are non-negative real numbers such that $x^2+y^2+z^2+x+2y+3z = \frac{13}{4}$,

(i) find the largest value of $x + y + z$;

(ii) prove that $x + y + z \geq \frac{\sqrt{22}-3}{2}$

Fundamental Rules

For any real numbers a, b and c:

1. (i) $a > b$ if and only if $a - b$ is positive,
   (ii) $a = b$ if and only if $a - b = 0$,
   (iii) $a < b$ if and only if $a - b$ is negative.
2. (i) If $a/b > 1$ and $b > 0$ then $a > b$,
   (ii) If $a/b = 1$ and $b > 0$ then $a = b$,
   (iii) If $a/b < 1$ and $b > 0$ then $a < b$.
   Ex. 9: Which is larger, 16^{18} or 18^{16}?.
3. If $a > b$ and $b > c$ then $a > c$. (Transitive property 傳遞性)
4. If $a > b$ then $a + c > b + c$. (Here c may be positive or negative.)

5. (i) If $a > b$ and $c > 0$ then $ac > bc$,
(ii) If $a > b$ and $c < 0$ then $ac < bc$.

6. If $a > b$ and $c > d$ then $a + c > b + d$.
(Note: The converse (逆定理) is not necessarily true.)

7. If $a > b > 0$ and $c > d > 0$ then $ac > bd$.
(Note: The converse (逆定理) is not necessarily true.)

8. (i) If $a > b > 0$ and $m > 0$, then $\frac{a}{b} > \frac{a+m}{b+m}$

(ii) If $b > a > 0$ and $m > 0$, then $\frac{a}{b} < \frac{a+m}{b+m}$.

(iii) If $\frac{a}{b} < \frac{c}{d}$, then $\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}$.

Ex.10 : If $\frac{1}{2} < a < 1$ and $\frac{1}{2} < b < 1$, prove that $\frac{1}{a+1} + \frac{1}{b+1} < \frac{3}{2}$.

Ex. 11：Determine all possible values of $\frac{a}{a+b+d} + \frac{b}{b+c+a} + \frac{c}{c+d+b} + \frac{d}{d+a+c}$
where a, b, c, d are arbitrary positive real numbers. (1974 IMO No. 5):

9. If $a > b > 0$ and $n > 0$ then $a^n > b^n$, and hence $a^{-n} < b^{-n}$.

Note: Consider the case $n=\frac{1}{2}$, the theorem means $a > b > 0 \Rightarrow \sqrt{a} > \sqrt{b}$.
In other words, provided $a>0$ and $b>0$, $a>b \Leftrightarrow a^2 > b^2$.

Ex. 12: Prove that $4 + \sqrt{7} < 2\sqrt{3} + \sqrt{11}$

10. (i) If $ab > 0$ then either ($a > 0$ and $b > 0$) or ($a < 0$ and $b < 0$),
(ii) If $ab < 0$ then either ($a > 0$ and $b < 0$) or ($a < 0$ and $b > 0$).

11. For any real numbers a, $a^2 \geqq 0$.
(Note that $a^2 \geqq 0$ does not imply that $a \geqq 0$.)

Ex. 12a: Prove that for any positiveve real number x, $x + \frac{1}{x} \geqq 2$.
Hence prove that for any real number a, b, c, $\frac{b+c-a}{a} + \frac{c+a-b}{b} + \frac{a+b-c}{c} \geqq 3$

12. The quadratic equation $ax^2 + bx + c = 0$, where $a \neq 0$, gives real roots in x if and only if $b^2 - 4ac \geqq 0$.

13. The function $f(x) = ax^2 + bx + c$, where $a \neq 0$, assumes positive values for all values of x if and only if $a > 0$ and $b^2 - 4ac < 0$.

14. $|a| \leqq |b|$ if and only if $-|b| \leqq a \leqq |b|$.

15. $||a| - |b|| \leqq |a + b| \leqq |a| + |b|$.

16. $|a_1 + a_2 + a_3 + \ldots\ldots + a_n| \leqq |a_1| + |a_2| + |a_3| + \ldots\ldots + |a_n|$.

Methods used in proving absolute inequalities:

1. Direct Comparison (比較法)
   (Use subtraction or division.)

   Ex. 13: Given that a>0 and b>0, prove that $\frac{a^2}{b}+\frac{b^2}{a}\geq$ a+b. (Hint: use subtraction)

   Ex. 14: If no two of the three positive real numbers a, b and c are equal, prove that (i) $a^ab^b>a^bb^a$. (ii) $a^ab^bc^c>a^cb^ac^b$. (Hint: use division)

   Ex. 15 If a, b, c are distinct positive real numbers, prove that $a^ab^bc^c > (abc)^{\frac{a+b+c}{3}}$.

2. Grouping Method (綜合法)
   (Group terms together, and use Method of Completing Squares.)
   Ex. 16: Let a, b, c be the sides of a triangle, and T its area.
   Prove: $a^2 + b^2 + c^2 \geq 4\sqrt{3}$ T. In what case does the equality hold?
   (Third IMO (1961) No. 2)

   Ex. 17: Given a, b and c are real numbers. Prove that
   $\sqrt{a^2+b^2}+\sqrt{b^2+c^2}+\sqrt{c^2+a^2}\geq\sqrt{2}$ (a+b+c)

3. Backward Trace (反推法) (Note: Ex. 12 is a typical example.)
   (Working backwards, should make sure that the symbol ⇔ is valid.)
   Ex. 18: Given that a, b are real numbers and a>b, prove that $\frac{a}{2}\geq\sqrt{2a(a-b)}-(a-b)$

   Ex. 19. Let $a_1, a_2, \ldots, a_n$ be real numbers greater than 1. Furthermore, for any k such that $1 \leq k \leq n-1$, $|a_{k+1} - a_k|<1$. Prove that $\frac{a_1}{a_2}+\frac{a_2}{a_3}+\ldots+\frac{a_{n-1}}{a_n}+\frac{a_n}{a_1}<2n-1$.

4. Telescoping Method (放縮法)
   (Used when dealing with sum of series.)
   Ex. 20: For any positive integer n, prove that
   $$\frac{3n+1}{2n+2}<1+\frac{1}{2^2}+\frac{1}{3^2}+\ldots+\frac{1}{n^2}<\frac{7}{4}.$$

   Ex. 21: Prove that $16<\sum_{k=1}^{80}\frac{1}{\sqrt{k}}<17$.

5. Use of Contrapositive Inference (反証法)
   (Assume the truth of a statement which is opposite to what is required, and then prove that it leads to contradiction.)
   (i.e., instead of proving P => Q, we try to prove that ~Q => ~P.)

   Ex. 22: Given that a, b, and c are real numbers such that abc > 0, a+b+c > 0 and ab+bc+ca > 0, prove that a, b and c are all positive.

   Ex. 23: Prove that if a and b are real numbers such that $a^3 + b^3 = 2$, then $a+b \leq 2$.

   Ex. 24: Given that a, b, c are real numbers. For any real number x, real numbers A, B, C, D satisfy $(Ax + B)(Cx + D) = ax^2 + bx + c$. Prove that among a, b, c, there is at least one which is greater than or equal to $\frac{4}{9}$(A+B)(C+D).

6. Change of Variable (換元法)
   (Expressing the inequality in terms of a new set of variables might give hints leading to method of solving the problem.)

   Ex. 25: Suppose a, b, a are the sides of a triangle, prove that
   $a^2(b+c-a) + b^2(c+a-b) + c^2(a+b-c) \le 3abc$. (Sixth IMO (1964) No. 2)

   Ex. 26: Given that $a^2 + b^2 = 1$, prove that $(a^2 + \frac{1}{a^2})(b^2 + \frac{1}{b^2}) \ge \frac{25}{4}$.

   Ex. 27: Given that a, b, c are real numbers such that $a \ge b \ge c \ge 0$, and $a + b + c = 3$,
   prove that $ab^2 + bc^2 + ca^2 \le \frac{27}{8}$. When does the equality hold?

7. Use of Discriminant (判別式法)
   (The roots of a quadratic equation are real if and only if $\Delta \ge 0$.)

   Ex. 29: Prove that for any real number a, $\frac{1}{2} \le \frac{a^2+a+1}{a^2+1} \le \frac{3}{2}$.

8. Mathematical Induction (數學歸納法)
   (Use in proving inequalities involving a variable n, a positive integer.)

   Ex. 30: For any positive integer $n \ge 5$, prove that $2^n > n^2$.

   Ex. 31: If $x_1, x_2, \ldots, x_n$ are all real numbers in the interval (0, 1) and their sum is equal to S_n, prove that for $n>2$, $(1 + x_1)(1 + x_2)\ldots(1 + x_n) > 1 + S_n$.

   Ex. 32: Given that n is a positive integer, prove that $1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{2^n} > \frac{n}{2}$.

9. Function Analysis (函數分析法)
   (Attempt to sketch the graph of a function.)

   Ex. 33: Find the minimum value of (i) $f(x) = \sqrt{x^2 - 3} + \frac{1}{\sqrt{x^2-3}}$

   (ii) $f(x) = \sqrt{x^2 + 3} + \frac{1}{\sqrt{x^2+3}}$

   Ex. 34: Given that $0<x<1$, $0<y<1$, $0<z<1$, prove that $x(1-y) + y(1-z) + z(1-x) <1$.